

Reverse Engineering of PixMob Cement V1.1r3 Wristbands

Daniel Nikolaev

Introduction

The University of Georgia (UGA) Athletic Association is always looking for ways to improve the fan experience at football games. One such way they underwent this was to provide light-up wristbands (built by PixMob) at two of the football games during the 2025 season.

- UGA vs the Alabama Crimson Tide on September 27
- UGA vs the Texas Longhorns on November 15

Taking a quick look at the wristbands, they seem identical. However, closer inspection leads to the discovery that they are from different batches. When broken open, the internals are identical.

PixMob Wristband Lot Numbers Internal PCB Version

vs Alabama: V1-2407-4305	vs Alabama: Cement V1.1r3
vs Texas: VN01-2505-5111	vs Texas: Cement V1.1r3

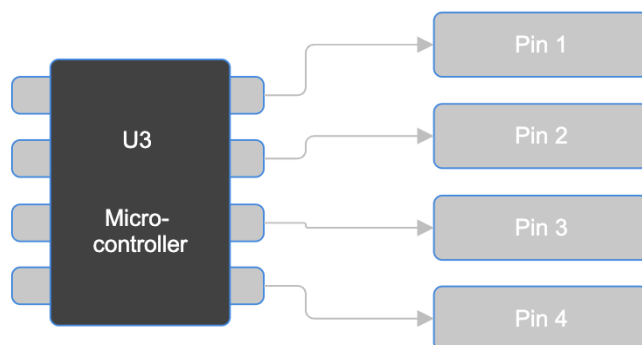
Physical Breakdown and Reverse Engineering of the Internal PCB

FIGURE 1 - Contents of the Wristbands



Internally, the wristbands from the two games are identical. Simply put, they contain [figure 1] a Printed Circuit Board (PCB), two CR2032 “button” batteries, a small strip of metal, and two Light-Emitting Diodes (LEDs).

FIGURE 2 - U3 Microcontroller Schematic



The U3 microcontroller on the PCB seems to send the signals necessary to the LEDs in order to produce color. In order to activate the LEDs - regardless of color- Pin 1 needs to be powered. Pins 2-4 correspond to the 3 colors used in computing to create color. Red, Green, Blue - RGB. Connecting (or “shorting”) Pin 1 and Pin 2 causes a crisp, blue light. Shorting pins 1-4 causes a bright, red light. I was unable to get what I consider a “clean connection” for pins 1-3, but by deductive reasoning, I can understand that shorting pins 1-3 leads to a green color.

Electronic Reverse Engineering using Radio Frequency (RF) /Infrared (IR)

Currently, I am unable to conduct experiments to wirelessly send lighting commands to the wristbands, in a similar fashion to what occurs during events using this technology.

However, I have discovered that the Cement V1.1r3 PCB uses radio frequency (RF) in order to receive signals. More specifically, the frequency of 915MHz. In the future, I should be able to use a Raspberry Pi in order to output commands using RF.